



HomeAir

DESIGN DOSSIER

Smart Indoor Air Quality Monitor

A complete design record documenting the full development lifecycle of HomeAir — from problem discovery and concept ideation through prototype iterations to the final integrated system.

ESP32

IoT Sensors

Mobile App

Alexa AI

TEAM MEMBERS

Project roles and responsibilities

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Khalid Dallol

ELECTRICAL ENGINEER STUDENT

Circuit development, app, backend, dashboard, Alexa.



Michel El Khalil

ELECTRICAL ENGINEER STUDENT

Dossier Design, system documentation, integration support.



Piero Garcia

ELECTRICAL ENGINEER STUDENT

Hardware design, sensors, circuit development, prototype.



Jonathan Pedro

ELECTRICAL ENGINEER STUDENT

Video production, firmware support, testing, integration.

DESIGN PROCESS

Double Diamond methodology applied to HomeAir

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01



DISCOVER

Research into indoor air quality problems. User needs analysis and environment assessment.

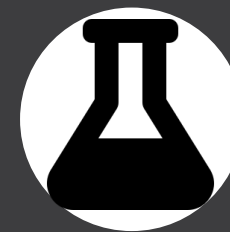
02



DEFINE

Problem statement formulation. Identifying key requirements and system constraints.

03



DEVELOP

Concept ideation, hardware prototyping, software development and iterative testing.

04



DELIVER

Final system integration, user testing, enclosure design and project documentation.

PROBLEM STATEMENT

Identifying the need for HomeAir

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- According to **Holden (2023)**, 22% of the modifiable factors contributing to child deaths in England from April 2019 to March 2020 were due to the child's physical environment, including overcrowding and unsuitable housing (including the presence of damp, mould and being in a state of disrepair)
- As one prominent example, the World Health Organization estimates 3.8 million people worldwide die every year from illnesses attributable to harmful indoor air from dirty cookstoves and fuel (**National Institute of Environmental Health Sciences, 2024**).
- On the 16th of December 2020 legal history was made when a ruling was made that air pollution was the cause of the death of a nine-year-old girl, Ella KissiDebrah (**Bennett, 2021**).
- In the UK, damp and mould caused by inadequate airflow are particularly prevalent, contributing to conditions like bronchitis and sinus infections (**VENTI Group Ltd, 2023**).
- While basic air quality monitoring devices are relatively inexpensive, advanced sensors that provide higher precision and additional features can be cost-prohibitive for some users. This price barrier can limit access for lower-income households and communities that are often the most affected by poor air quality (**Sensor, 2025**).

Key issues identified:



No real-time CO₂ or air quality feedback in typical homes



Existing solutions are costly, complex, or not user-friendly



Poor air quality causes fatigue, headaches, and long-term health risks

PROJECT OBJECTIVES

What HomeAir is designed to achieve

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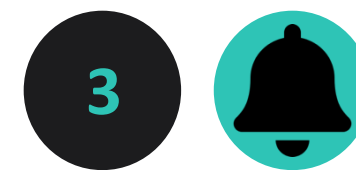
Monitor Air Quality

Continuously measure CO₂ concentration, temperature, humidity, and derive an overall air quality index in real time.



Accessible Dashboard

Provide a clear and user-friendly mobile interface displaying live sensor readings with intuitive visual indicators.



Proactive Alerts

Notify users immediately when environmental conditions exceed healthy thresholds, enabling timely corrective action.



Wireless Connectivity

Transmit sensor data reliably over Wi-Fi from the ESP32 to the mobile application without user intervention.



Promote Healthier Environments

Give users actionable insights and recommendations to improve indoor air conditions for wellbeing.



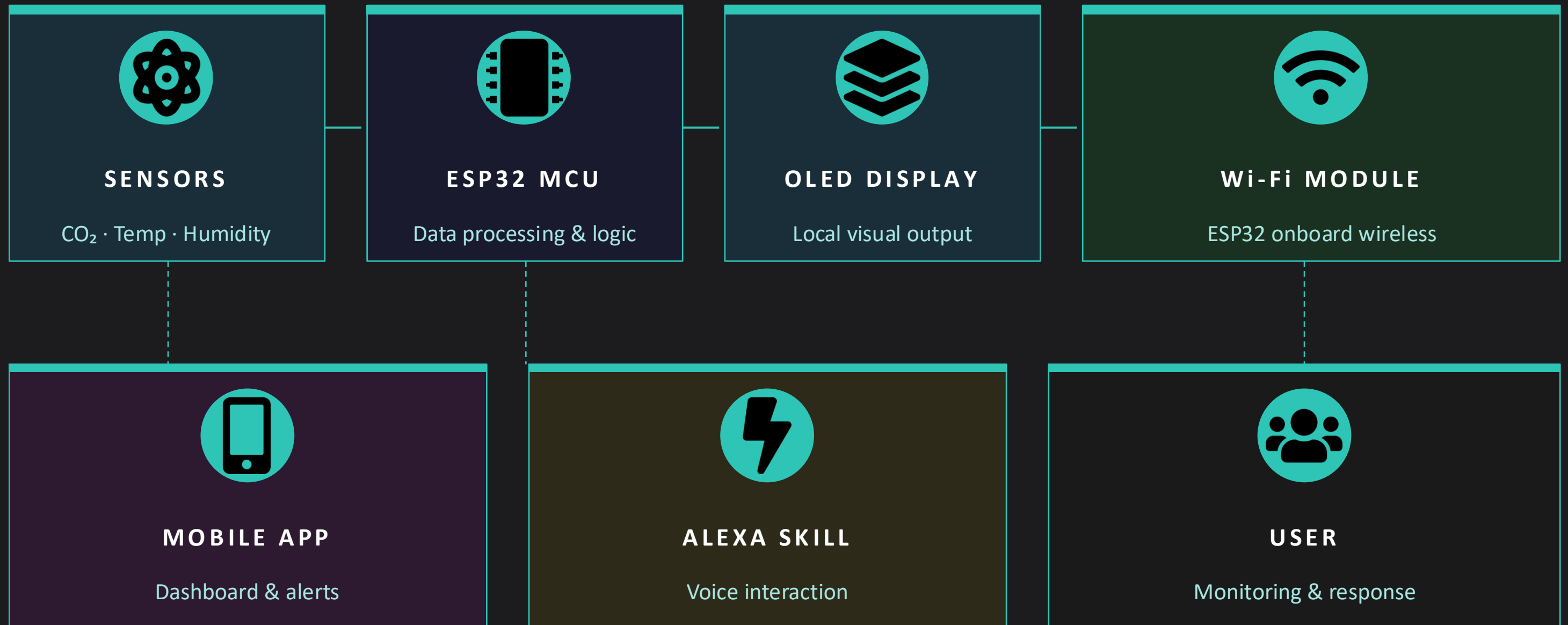
Affordability & Accessibility

Develop a low-cost and user-friendly solution suitable for everyday home use.

SYSTEM ARCHITECTURE

Component connectivity and data flow diagram

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Data Flow →

RISK REGISTER

Identified risks, likelihood, impact and mitigations

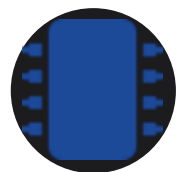
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RISK	LIKELIHOOD	IMPACT	LEVEL	MITIGATION STRATEGY
R1 Sensor inaccuracy or drift	Medium	High		Calibration checks; cross-reference with reference device
R2 ESP32 Wi-Fi connectivity failure	Medium	High		Reconnection logic in firmware; local OLED fallback display
R3 Power supply instability	Low	Medium		Use regulated USB supply; add capacitors to smooth supply
R4 Mobile app compatibility issues	Low	Medium		Test across Android versions; use cross-platform framework
R5 Alexa API deprecation or changes	Low	Low		Monitor AWS documentation; decouple Alexa layer from core
R6 Time constraints on features	High	Medium		Prioritise core system functionality to meet project deadlines

PROTOTYPE DEVELOPMENT

Iterative improvements from breadboard to final device

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1. Breadboard Assembly

WEEK 1-2

Initial wiring of all sensors to ESP32 on breadboard. Basic data acquisition confirmed via serial output.



2. Display Integration

WEEK 3

OLED display connected and programmed. Real-time CO₂ and temperature values rendered with clear formatting.



3. Wi-Fi & App Link

WEEK 4

ESP32 programmed to transmit JSON data packets over Wi-Fi. Mobile app configured to receive and render data.



4. Wiring Refinement

WEEK 5

Breadboard replaced with cleaner layout. Component connections organised to reduce noise and improve reliability.



5. Enclosure Design

WEEK 6

Components placed inside custom casing. Sensor placement optimised for accurate ambient readings.



6. Testing & Iteration

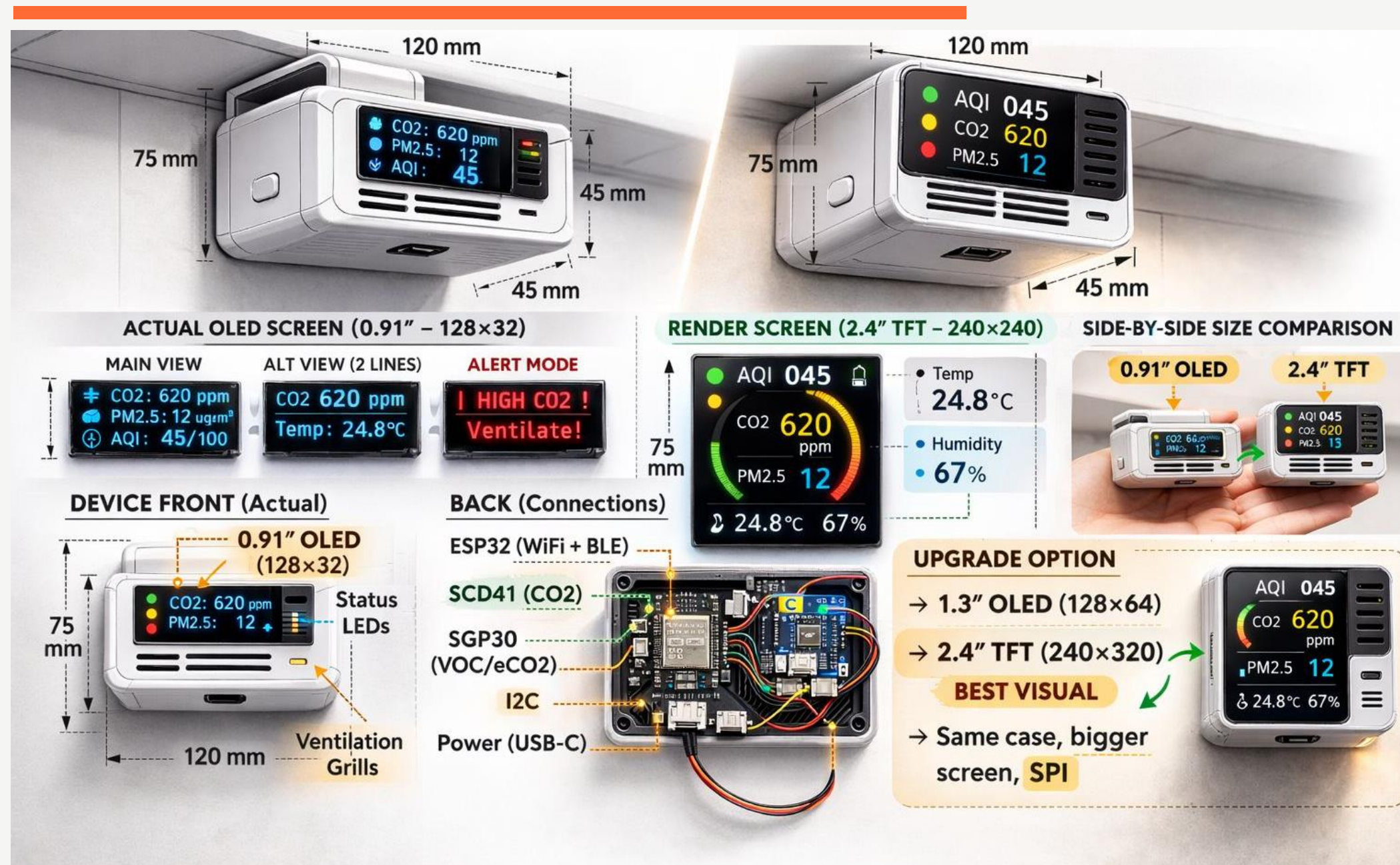
WEEK 7-8

Full system tested. Alert thresholds tuned. Alexa integration added. Concept product render produced.

INITIAL PROTOTYPE

Phase 1 — Breadboard prototype and component integration

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- The figure presents the development of the initial prototype and its progression towards the final device design.

- The breadboard implementation integrates the ESP32 microcontroller with CO₂, temperature, and humidity sensors, along with an OLED display for real-time monitoring.

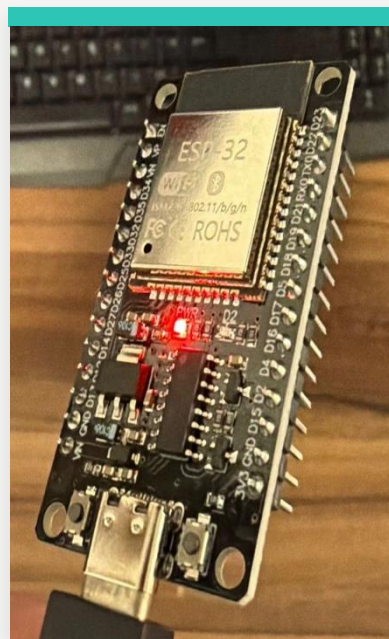
- A comparison with the conceptual enclosure highlights key design improvements, including a larger display and enhanced user interface.

- This stage was used to validate system functionality and sensor performance before refining the final product.

MATERIAL SELECTION

Components, sensors and physical materials chosen for HomeAir

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MICROCONTROLLER

ESP32

Cost: £6.99

→ High processing power, built-in wireless, low cost



AIR QUALITY

MQ-135 Sensor

Cost: £5.49

→ Wide gas detection range, analog output, low cost

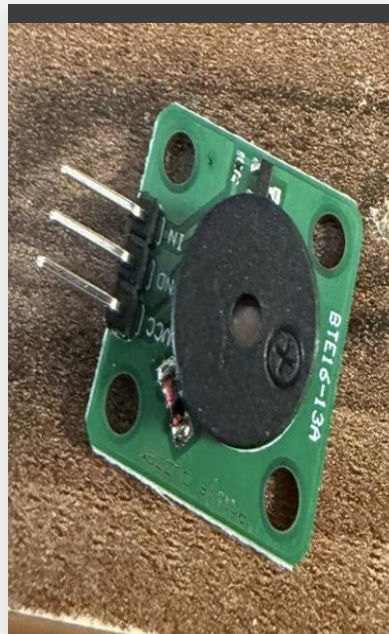


AIR QUALITY

SGP30 Sensor

Cost: £10

→ VOC and air quality measurement



ALERTS

Buzzer

Cost: £8.99

→ Audio Alert for poor air quality



Display

OLED Screen

Cost: £5.99

→ Displays real-time air quality data



ALERTS

LEDs(Red, Yellow and Green)

Cost: £5.99 (60 pieces)

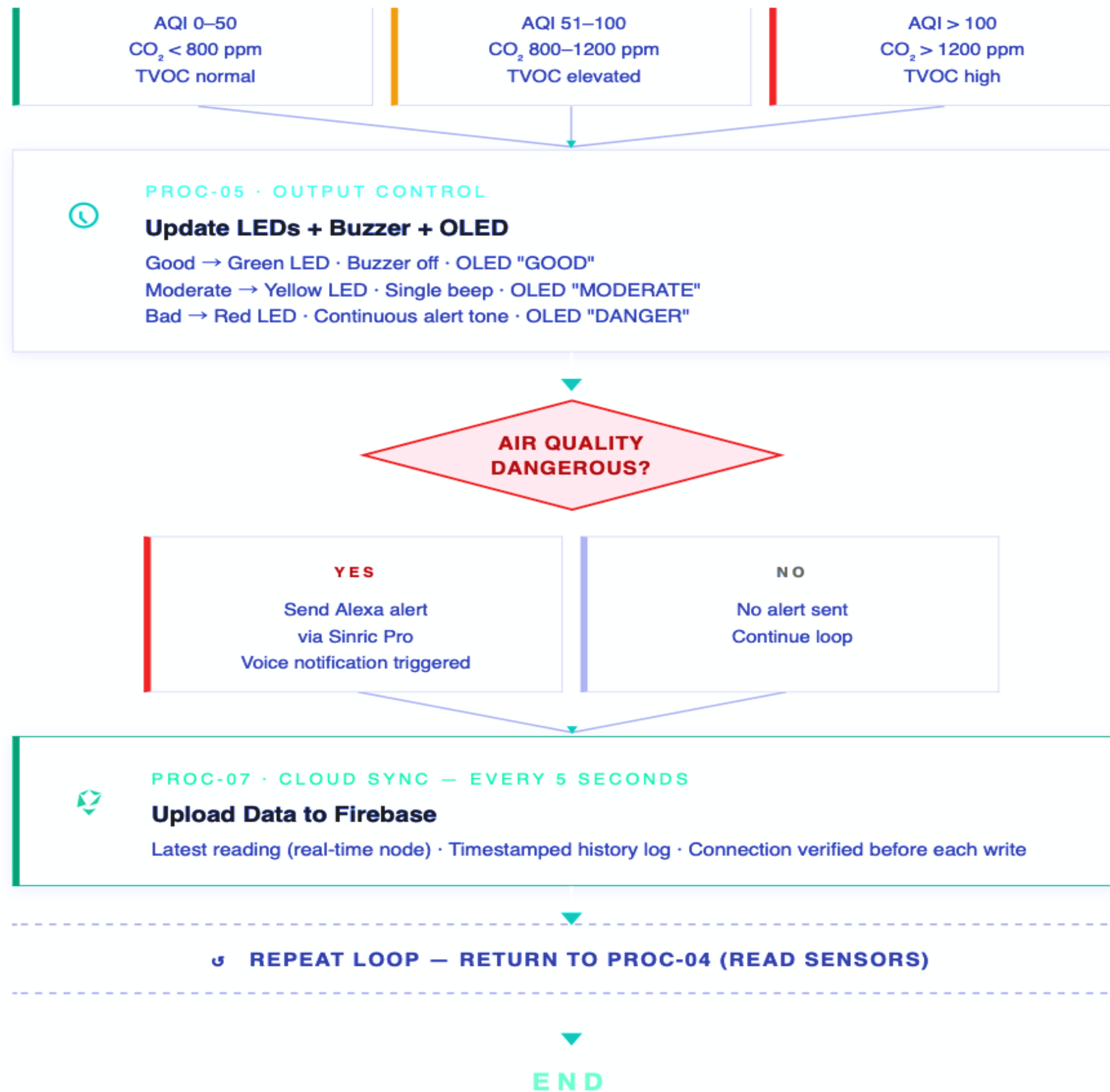
→ Visual air quality indication

SYSTEM OPERATION FLOWCHART OF HOMEAIR DEVICE



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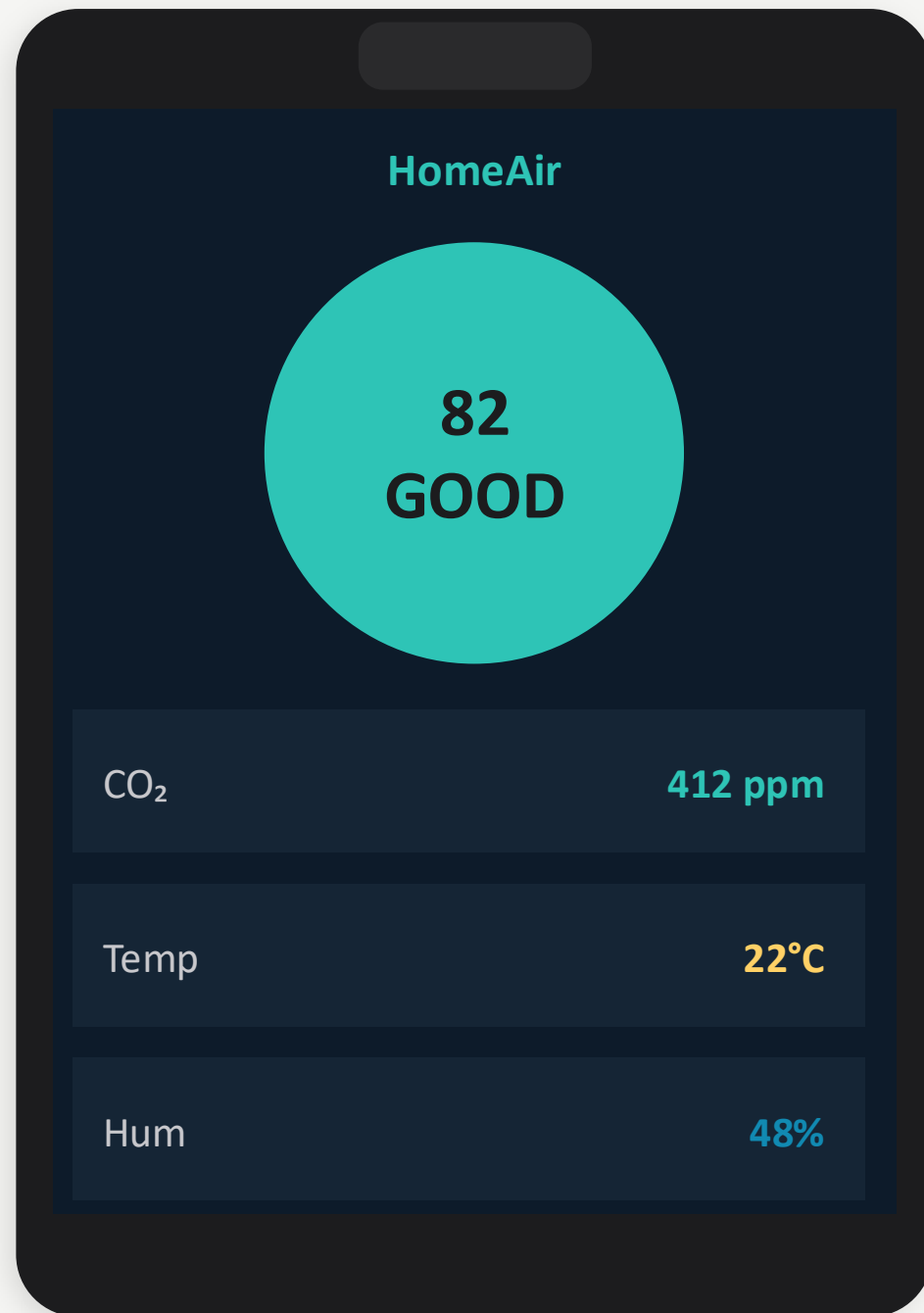




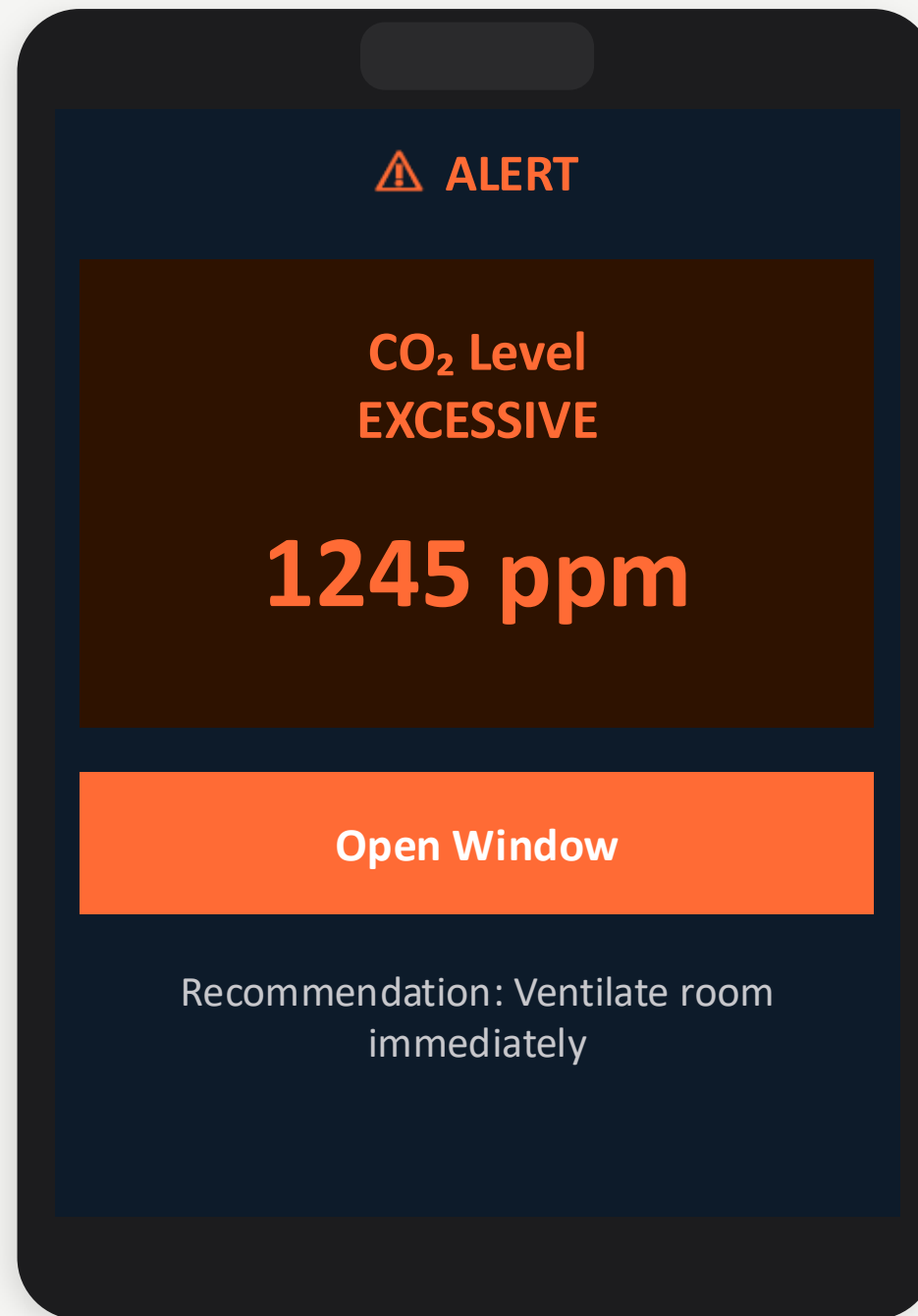
APPLICATION DESIGN

Mobile UI sketch — dashboard interface and user flows

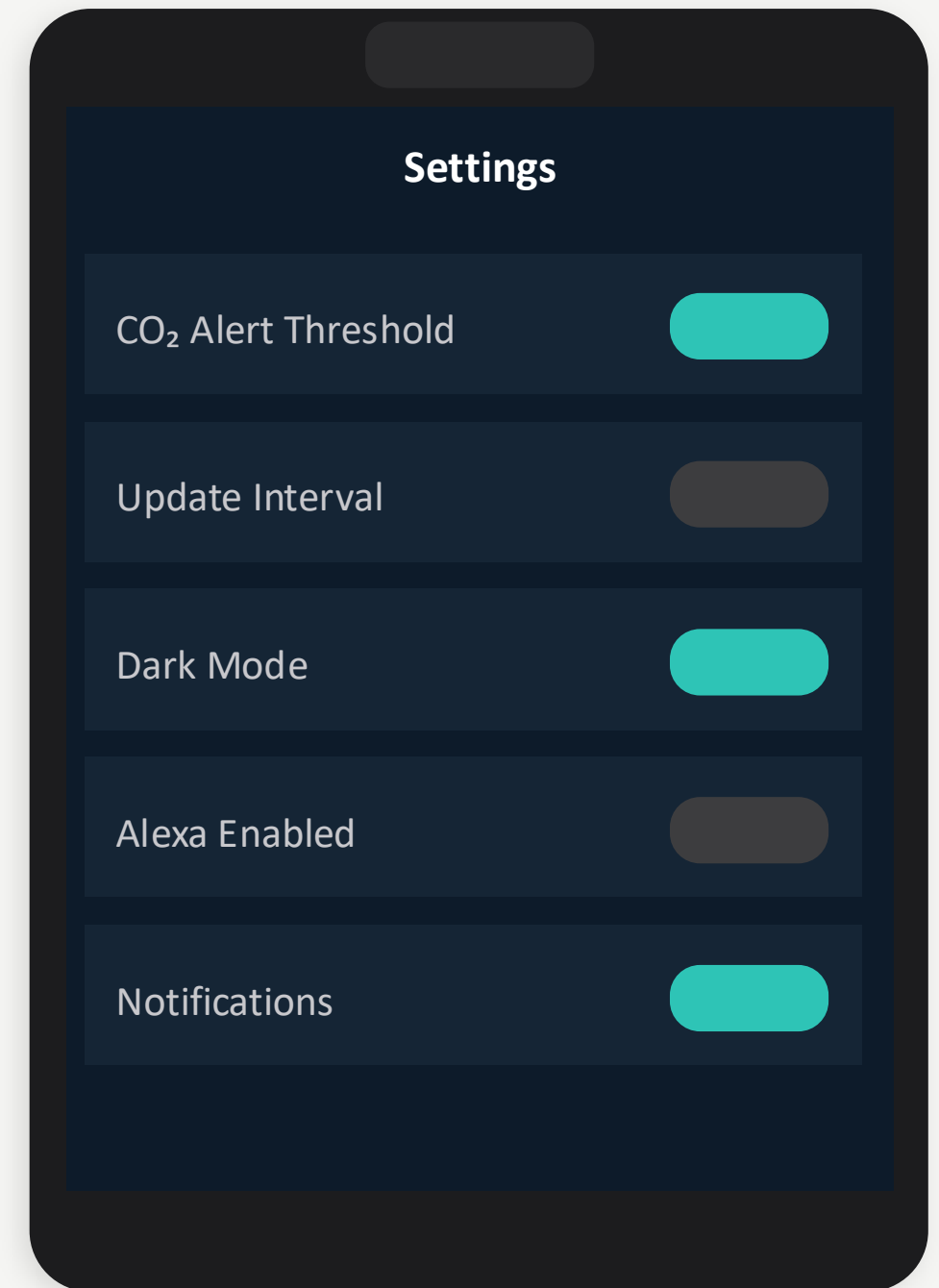
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Home Dashboard



Alert View

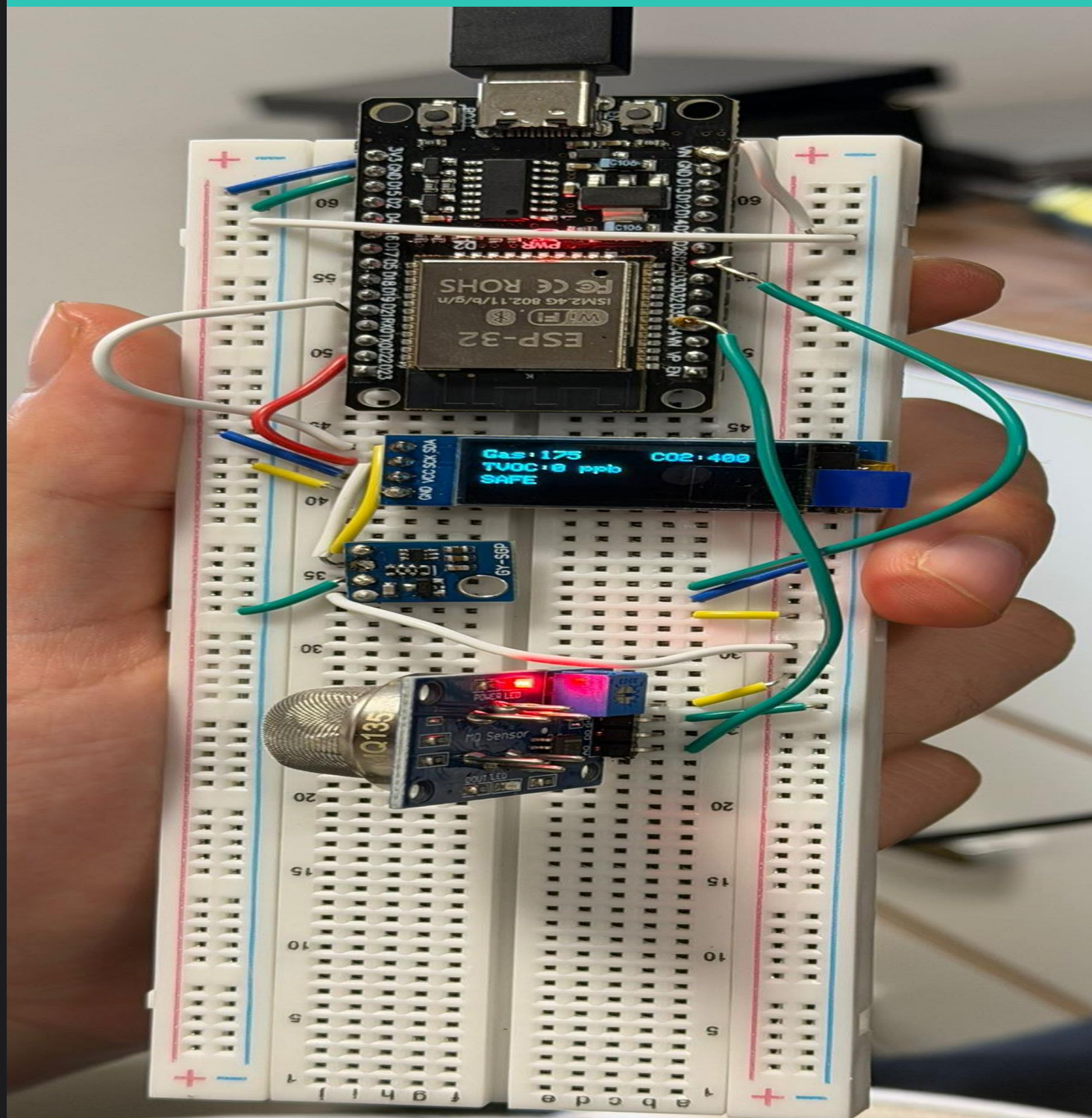


Settings

SYSTEM IMPLEMENTATION

Hardware implementation of the HomeAir System UI sketch

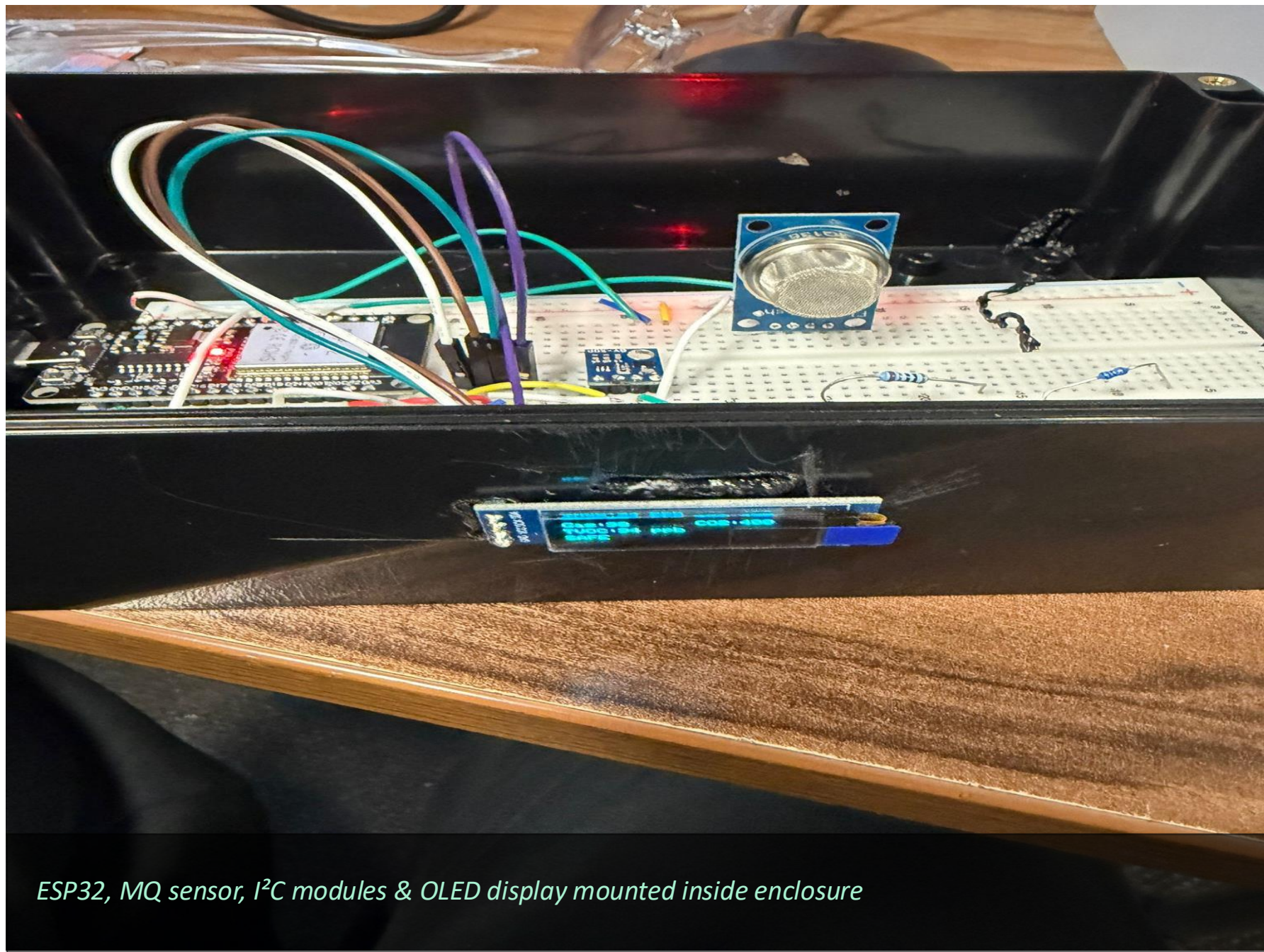
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The HomeAir system was implemented using a breadboard-based prototype built around the ESP32 microcontroller. During this stage, air quality sensors, including CO2 and environmental sensing modules, were integrated and connected to enable real-time data acquisition. The system was programmed to process sensor readings and display key parameters such as gas concentration and environmental conditions on a compact OLED screen. This allowed continuous monitoring and immediate visual feedback during testing. The prototype setup provided flexibility for adjusting connections, testing different configurations, and troubleshooting issues. Overall, this stage was essential for validating sensor integration, communication, and system performance before progressing to the final device design and enclosure development.

INTERNAL ASSEMBLY

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ABS Project Enclosure

Black project box houses all components securely. Lid left open showing internal layout and wiring organisation.

MQ Gas Sensor

Blue sensor module mounted on breadboard — red LED status indicators visible in upper section of enclosure.

Breakout Modules

Small blue sensor modules on the breadboard handle I²C communication for temperature, humidity, and display.

OLED Display

SSD1306 screen fitted through slot in the enclosure front — displaying AQI, CO₂ and SAFE status live.

ESP32, MQ sensor, I²C modules & OLED display mounted inside enclosure

SYSTEM RESULTS AND LIVE TESTING

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100

/100

AQI Score

Air quality rated GOOD

400

ppm

CO₂ Level

Within safe threshold

373

raw

Gas Reading

GOOD — no VOC concern

3

ppb

TVOC Level

Excellent — very low

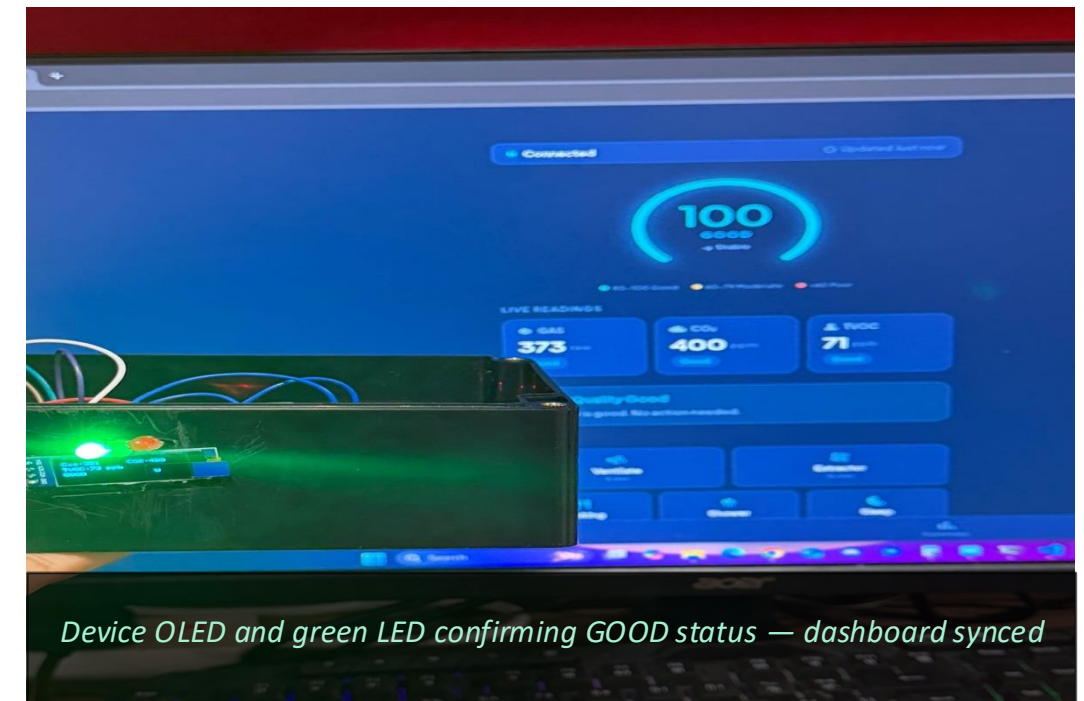
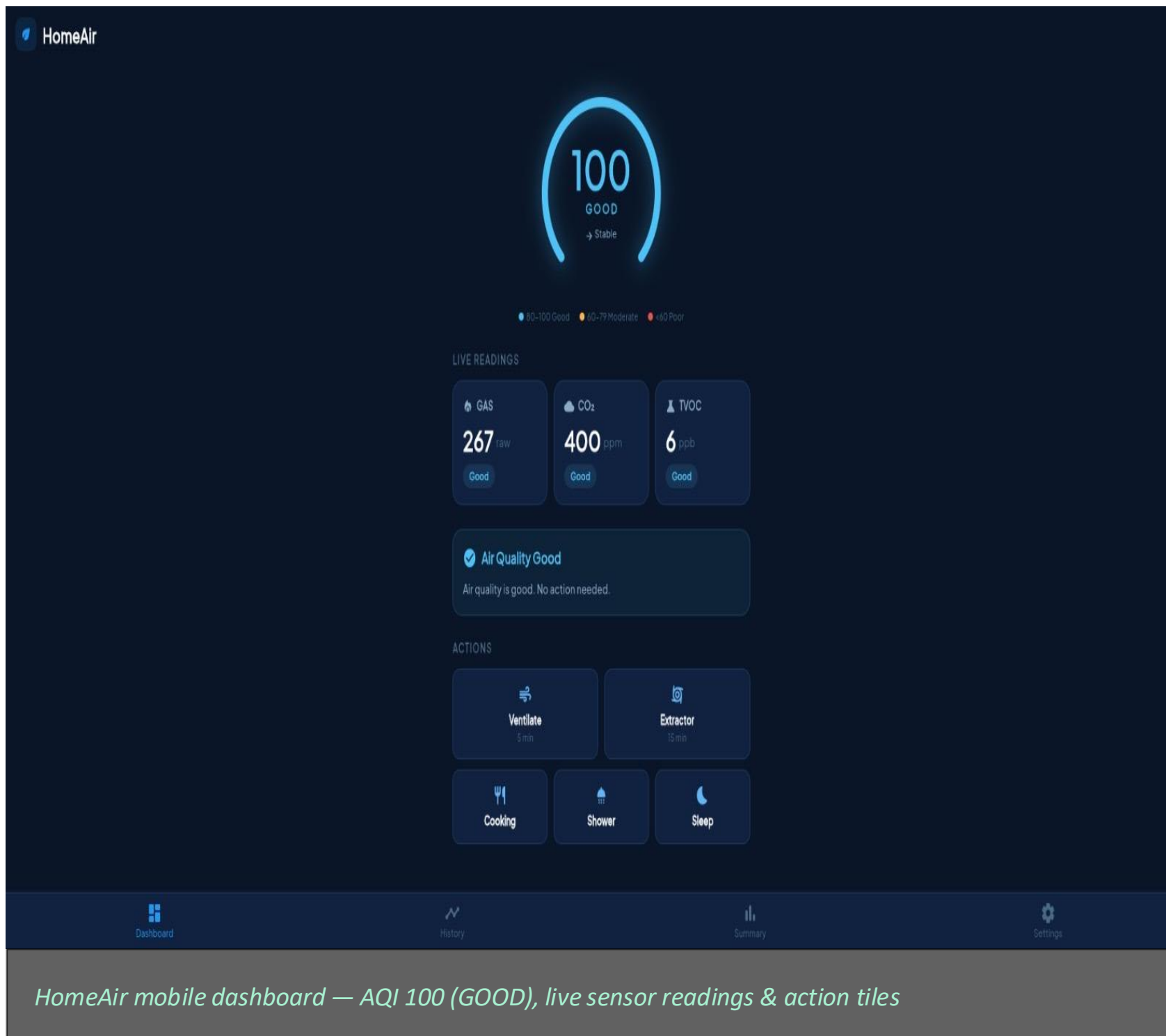
HomeAir device (lid open) alongside Amazon Echo — live integration confirmed

CONNECTED · LIVE

All sensors operational. Data transmitting in real time to the HomeAir dashboard. System status confirmed GOOD across all parameters during live testing session.

DASHBOARD AND DEVICE SYNCHRONIZATION

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AQI 100 — GOOD

Composite score confirms excellent indoor air quality across all sensor channels.

Wireless Sync Confirmed

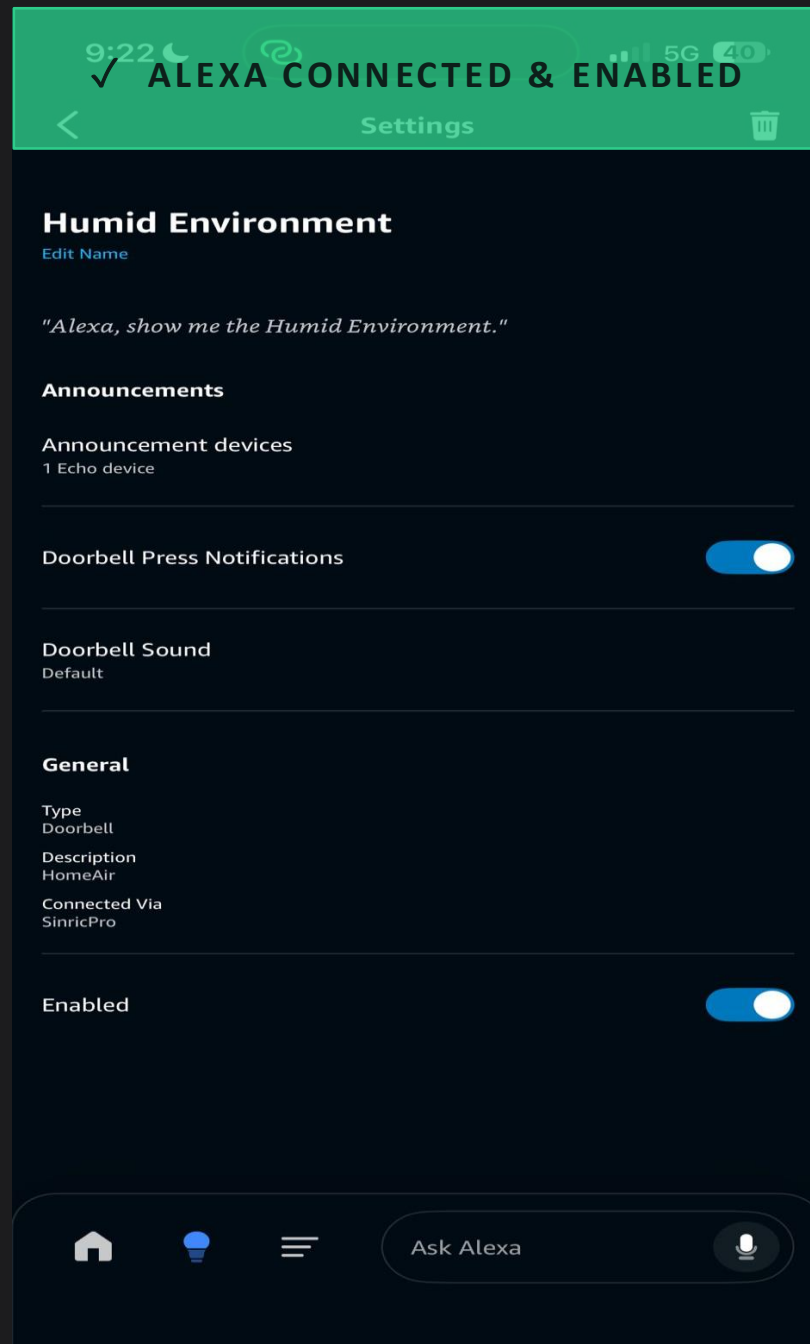
Device transmits live data to dashboard in real time — CONNECTED status shown.

Action Recommendations

Dashboard surfaces contextual actions (Ventilate, Extractor, Sleep) based on readings.

VOICE INTEGRATION . AMAZON ALEXA . VERIFIED OPERATIONAL

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01 Device Registered in Alexa App

HomeAir is registered as a device named "Humid Environment" in the Alexa ecosystem — visible in Settings with full configuration controls.

Connected via SinricPro

03 Echo Device Linked

1 Echo device is registered and paired under Announcement Devices — confirming the physical Amazon Echo unit is active and receiving data from HomeAir.

1 Echo Device Active

02 Voice Command Functional

The Alexa voice prompt "Alexa, show me the Humid Environment" is configured and operational — enabling hands-free air quality queries.

"Alexa, show me the Humid Environment."

04 Device Status: Enabled

The Enabled toggle is ON in the Alexa app. Doorbell Press Notifications are active, allowing HomeAir to push alerts through the Alexa speaker when thresholds are exceeded.

Enabled + Notifications ON

✓ Alexa integration verified as fully operational — device detected, voice commands active, Echo device linked, notifications enabled

FEATURES & INNOVATION

What makes HomeAir distinctive

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Multi-Parameter Sensing

Simultaneous CO₂, temperature and humidity monitoring for comprehensive air quality assessment.



Composite AQI Score

Proprietary algorithm combines all sensor readings into a single, intuitive 0–100 air quality score.



Live Mobile Dashboard

Real-time data visualisation with colour-coded indicators and push notifications for poor conditions.



Alexa Voice Integration

Users can query air quality conditions by voice, making monitoring effortless and hands-free.

Innovation Highlights



Single-device solution combining sensing, display, wireless and voice in one low-cost unit



Threshold-based alert engine with user-configurable limits for personalised safety



Architecture designed for extensibility — new sensors and integrations can be added modularly



Concept render produced to visualise the system as a finished commercial product

LIMITATIONS & IMPROVEMENTS

Current constraints and future development roadmap

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CURRENT LIMITATIONS

-  Alexa integration reliability inconsistent under variable network conditions
-  Application not yet available on official iOS/Android app stores
-  No on-device audio/buzzer alert — relies solely on mobile notifications
-  Enclosure lacks professional branding and final surface finish

PLANNED IMPROVEMENTS

-  Robust Alexa reconnection logic and faster voice response handling
-  Publish to Google Play and App Store with CI/CD pipeline
-  Integrate onboard buzzer for standalone audio alerts
-  Professional enclosure with HomeAir branding, logo and refined finish



CONCLUSION

- Working smart air quality monitor delivered
- Full IoT pipeline: sensors → MCU → app → voice
- Real-time data, alerts and AQI implemented
- Double Diamond process successfully applied
- Clear roadmap for future development identified

Project Outcomes

HomeAir successfully demonstrates how affordable embedded hardware, wireless connectivity, and an intuitive mobile interface can be integrated to deliver a meaningful real-world health monitoring solution. The project applied the Double Diamond design process rigorously, producing a functional prototype with a clear path to commercial development.

REFERENCES

Sources consulted during the HomeAir design process

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- [1] National Institute of Environmental Health Sciences (2024) *Indoor air quality*, National Institute of Environmental Health Sciences. Available at: <https://www.niehs.nih.gov/health/topics/agents/indoor-air>.
- [2] Bennett, P. (2021). *The importance of indoor air quality*. Energy in Buildings & Industry. Available at: [PDF].
- [3] VENTI Group Ltd (2023) *VENTI Group Ltd*. Available at: <https://www.venti-group.com/docs/what-health-issues-can-poor-ventilation-cause/>.
- [4] Holden, K.A. (2023) 'The impact of poor housing and indoor air quality on respiratory health in children', *Breathe*, 19(2). Available at: <https://doi.org/10.1183/20734735.0058-2023>.
- [5] Sensor, R. (2025) *What Are The Advantages & Disadvantages Of Air Quality Sensors?*, *Rikasensor.com*. Available at: <https://www.rikasensor.com/a-what-are-the-advantages-disadvantages-of-air-quality-sensors.html>.



HomeAir

Smart Indoor Air Quality Monitoring System

Real-time IoT sensing • Firebase cloud • Flutter app • Alexa voice alerts



Project Overview

HomeAir is a smart indoor air quality monitoring system that combines ESP32 IoT hardware, multi-sensor fusion, Firebase cloud infrastructure, and a Flutter mobile/web application.

It delivers real-time environmental data, actionable intervention recommendations, mould risk analysis, and weekly air quality insights.

Our thesis: UK councils should mandate installation of affordable air quality monitors in residential properties to protect public health and prevent property damage.

Hardware Components



ESP32 Dev Module

WiFi-enabled microcontroller handling sensor acquisition, cloud upload, Alexa integration, and LED/buzzer control.

MQ-135 Gas Sensor

Analog sensor on GPIO34 detecting ammonia, benzene, CO₂, and smoke. Provides raw pollution spike data.

SGP30 VOC Sensor

I²C digital sensor delivering calibrated eCO₂ (ppm) and TVOC (ppb) readings. Shares I²C bus with OLED.

SSD1306 OLED Display

128x32 pixel I²C display showing real-time readings, status, and WiFi connection state.

Buzzer + LED Indicators

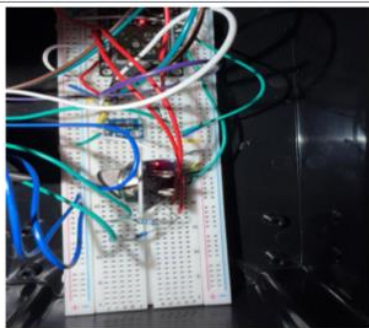
Active-low buzzer on GPIO25. Three LEDs: Green (GPIO17), Yellow (GPIO18), Red (GPIO19).

System Architecture

- ESP32 reads MQ-135 + SGP30 sensors every 1 second
- Data pushed to Firebase RTDB via HTTP every 5 seconds
- Flutter web app polls Firebase every 3 seconds
- Sinric Pro doorbell event triggers Alexa voice alert
- OLED + LEDs + buzzer provide local hardware feedback
- 60-second cooldown prevents repeated Alexa spam

Pin Mapping

MQ-135 AO → GPIO34 (analog)
SGP30 SDA → GPIO21 (I²C)
SGP30 SCL → GPIO22 (I²C)
OLED SDA → GPIO21 (shared bus)
OLED SCL → GPIO22 (shared bus)
Buzzer SIG GPIO25 (active-low)
Green LED → GPIO17
Yellow LED GPIO18
Red LED → GPIO19



Device Specifications

HomeAir operates through a cloud-connected IoT architecture. Environmental data is acquired by the ESP32 from connected sensors, including the MQ gas sensor and SGP30 digital air-quality sensor. The ESP32 processes these readings and transmits them via Wi-Fi to the mobile application and cloud backend, allowing live monitoring and analytics.

The Flutter app then presents the information through an intuitive dashboard, historical trends and intervention prompts.

When threshold conditions are exceeded, the system can trigger alerts locally and remotely. This architecture enables real-time monitoring, remote access and future smart-home expansion, including Alexa announcements through Sinric Pro integration.

As a result, the system supports both immediate user awareness and longer-term indoor environmental management.

Application Features

The HomeAir Flutter application runs on web (Chrome), Android, and iOS. It connects to Firebase Realtime Database via REST API and presents live sensor data through a polished, responsive interface.

- Real-time Air Quality Score (0–100) with composite scoring from gas, CO₂, and TVOC data
- Live sensor cards displaying Gas (raw), CO₂ (ppm), and TVOC (ppb) with colour-coded status chips
- Colour scheme: Green = Good, Yellow = Moderate, Red = Poor
- Interactive history charts (fl_chart) with metric switching (Gas / CO₂ / TVOC) and time range filters (5m, 15m, 1h, 6h, 24h)
- Weekly summary with air quality breakdown bar, average/max statistics, and mould risk assessment
- Smart recommendation engine providing contextual ventilation advice
- Intervention action buttons: Ventilate (5 min timer), Extractor (15 min timer), plus event logging for Cooking, Shower, and Sleep
- Trend analysis showing improving, stable, or worsening air quality direction
- Settings screen with adjustable alert thresholds and notification controls
- Supports both light and dark display modes

App Screenshots

The Dashboard screen shows the Co₂ levels, as well as gas and humidity while showing how good the levels are, with 'GOOD', 'MODERATE' and 'BAD'

The History & Analytics screen transforms raw environmental readings into useful insight. Time-series plots are used to visualise how air quality changes throughout the day, allowing short pollution spikes and longer ventilation problems to be identified clearly. Summary tiles beneath the graph present key indicators such as minimum, maximum and average values, giving the user a quick understanding of room conditions without needing to interpret the full chart manually.

This feature supports trend-based decision making rather than one-off monitoring. By reviewing previous readings, the user can identify repeated patterns linked to daily activities such as cooking, sleeping or poor ventilation periods. In the full HomeAir system, these analytics help quantify exposure duration, highlight threshold exceedances and support intelligent recommendations for improving indoor air quality over time.



Alexa Voice Alerts



HomeAir integrates with Amazon Alexa via Sinric Pro, a free IoT cloud platform. The ESP32 registers as a virtual doorbell device on the Sinric Pro network.

When gas readings exceed 800, the ESP32 fires a doorbell event through Sinric Pro's WebSocket API. The Alexa Echo Dot, linked via the Sinric Pro skill, speaks a customisable warning aloud.

A 60-second cooldown prevents repeated alerts while conditions remain poor.

Air Quality Thresholds

Level	Gas	eCO ₂	LED	Alert
GOOD	0–500	<600 ppm	Green	None
MODERATE	500–800	600–1000	Yellow	None
BAD	800+	>1000 ppm	Red	Buzzer + Alexa

Cloud Backend

Firebase Realtime Database (free Spark plan) stores all sensor data. The ESP32 sends simple HTTP requests — no SDK required.

```
FUT
/rooms/room1/latest.json
POST
/rooms/room1/readings.json
```

Each payload contains: gasRaw, co2ppm, tvoc, airLevel, and a server timestamp. The Flutter app polls via HTTP GET every 3 seconds for live updates.

Technology Stack

Hardware	ESP32, MQ-135, SGP30, SSD1306 OLED
Firmware	Arduino C++, Adafruit + SinricPro
libs	Firebase Realtime Database (REST)
Cloud	
API	Flutter / Dart, Provider, fl_chart
App	
Voice	Sinric Pro + Amazon Alexa Echo Dot

Project Team

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